

Kavin Desi Valli

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TECHNICAL SKILLS

Languages: Rust, JavaScript, TypeScript, Python, SQL, Dart, C++

Frameworks: React, Next.js, Node.js, OpenTelemetry, Kafka, Protobuf, gRPC, Redis, PostgreSQL, Django

Other Tools: Docker, Git, OpenCV, AWS, Vercel, Cloudflare Workers

EXPERIENCE

Software Engineer Intern

San Francisco, CA

Vercel

Jan. 2026 – Apr. 2026

- Engineered a **dual-VM sandbox pooling system** that reduced infrastructure costs by **20×** while improving sandbox startup performance.
- Shipped the **v0 Slack Agent** to General Availability, contributing production-ready features and launch readiness
- Customized and integrated a **fork of VS Code's code-server** into v0 to enable a browser-native AI coding environment.
- Developed production features and performance improvements across the v0 platform in a fast-paced engineering team.

Software Engineer Intern

San Francisco, CA

Helicone AI (YC W'23) 

Sept. 2024 – Dec. 2024, May 2025 – Aug. 2025

- Shipped new features and bug fixes in a rapid startup environment, merging approximately **200 pull requests**.
- Built production features using **Node.js, Next.js, Rust, OpenTelemetry, Protobuf, HDBSCAN clustering, and Kafka**.
- Created and maintained NPM, Python, and Go packages including [@helicone/async](https://github.com/helicone/async), [@helicone/helpers](https://github.com/helicone/helpers), [helicone-async](https://github.com/helicone-async), [helicone-helpers](https://github.com/helicone-helpers), and [go-helicone-helpers](https://github.com/go-helicone-helpers).
- Implemented features and reliability improvements in Helicone's Rust-based [AI Gateway](#).

Embedded Full Stack Developer

Toronto, Canada

Arcturus Networks Inc.

Jan. 2024 – Apr. 2024

- Independently developed the Brinq Edge AI platform for the **NXP i.MX 8M Plus** using **React, Node.js, and C++**, with development and testing performed on a **Jetson Xavier NX**.
- Integrated **YOLOv8 object detection, occupancy analytics**, and video clip saving using **FFmpeg**.
- Developed an **OpenCV** heatmap visualizing crowd intensity from real-time **HLS video streams**, showcased at **Embedded World 2024** on an **i.MX 93 SoM**.
- Reduced initial load times of **10–15 seconds**, optimized video storage using a circular buffer, and maintained deployment and **Jenkins pipelines**.

Lead Web Developer

Remote

Starmorph AI

May 2023 – Sept. 2023

- Implemented user authentication using **Supabase Auth**, rate limiting with **Redis**, and payment processing through **Stripe**.
- Developed an enhanced generative chatbot using **Next.js, LangChain, and OpenAI**.

PROJECTS

Pipelined Matrix-Vector Accelerator | *SystemVerilog, Vivado, FPGA*

- Designed an **8-lane pipelined matrix-vector accelerator** in SystemVerilog for an AMD Kria KV260 FPGA, achieving **~400 MHz** post-implementation using BRAM-backed memories and streaming valid/ready interfaces.

QR Code-Based Entry System | *TypeScript, Python, PHP, Dart, Laravel, React, Flutter*

- Built a multiplatform application for tracking attendance across **2,000+ attendees** at DPS R.K. Puram's 50th-anniversary celebrations. 

Typewind | *TypeScript, Rust, Next.js*

- Developer and maintainer of an open-source project with **2.1K+ GitHub stars**, combining TypeScript's type safety with the versatility of Tailwind CSS. 

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

Sept. 2023 – Apr. 2028

ACHIEVEMENTS

NASA Space Apps Global Finalist Honorable Mention | *NASA Earth Science Division*

Nov. 2022

- Ranked among the top **80 of 50,000+ teams** for building [Liberty](#).